Amendment 7: review correction

2026-10-21 · frozen run `run-20261021T0056Z` · PR 2366

Outcome: strict kernel-log attribution and the R1/R2 prediction scores are unverified. All four processes reported allocation failure and SIGABRT. No completed cost comparison or speedup is established.

Retained observations

Cell / target	Elapsed seconds	Peak RSS (KiB)	Virtual cap (KiB)
X1 / 0	1915.9	60,667,848	64,250,012
X1 / 1	1873.7	60,713,840	64,250,012
X2 / 0	6011.1	62,919,616	64,250,008
X2 / 1	6024.0	62,849,712	64,250,008

The original Linux files label these units `kB`; one KiB is 1,024 bytes. The virtual caps are about 61.27 GiB. Timings and peak RSS are observations from the four archived processes, without a new performance run.

Why attribution is withheld

The historical runner suppressed failures when reading the kernel log. It could therefore write "no kernel OOM lines" after either a successful empty match or an unsuccessful read. The two retained sentinel files do not distinguish those cases. The corrected native analyzer labels capture UNVERIFIED and withholds strict R1/R2 attribution and prediction scores. The earlier completed cells' recorded verdicts are unchanged.

`ulimit -v` limits virtual address space. An allocation failure under that cap does not establish that the minimum resident working set exceeds 64 GB. Required virtual space, allocator behavior, and required resident memory remain distinct. This correction preserves the observed allocation failures and does not recast them as completed measurements.

Repaired runner and validation

The runner requires readable kernel logs before producer execution; retains full before/after captures and capture errors; preserves a producer's nonzero exit status; and refuses to reuse an existing evidence prefix. It no longer clears the host's diagnostic ring. The analyzer requires an explicit successful-capture marker and rejects recorded OOM events even when no "Killed process" line exists. Earlier OOM events in the retained full ring conservatively withhold attribution.

Two native regression tests pass. Offline shell controls cover successful capture, unreadable pre-flight logs, failed post-run capture, failed producer execution, retained OOM events, and existing evidence refusal. These controls use mock commands; no cloud host or research engine was launched. The exact example source compiled against retained dependency artifacts; this is not a fresh full-library test claim.

Evidence

The frozen raw files, original analysis, and original report remain unchanged. This directory contains `analysis-amendment7-corrected.txt`, regenerated by the native analyzer from the same records, and `provenance.json` with their input hashes. The historical report is superseded only for the attribution and resident-memory claims described above.